Matthew Choi

matthew.choi02@gmail.com | (952) 913-1087 | Seattle, WA
linkedin.com/in/MattyChoi | github.com/MattyChoi | mattychoi.com

EXPERIENCE

Meta; *Software Engineer, Machine Learning* — Aug 2024 - Mar 2026

- **Built and shipped an ML classifier for detecting compromised high-profile Facebook Pages,** recovering **10x** more pages and doubling precision (**31%** → **70%**) while achieving daily detection over the previous strategy's cadence of 3 days.
- **Prototyped an LLM-based bulk inference pipeline** that cut scaled reviewer workload by **30%** at **99%** precision for page compromise cases, informing the team's production review strategy.
- **Led the end-to-end redesign of a system that proactively queues content-creator impersonator accounts for takedown,** combining ML signals and heuristics into a unified priority-ranking model.
  - Increased action rate (precision) **2.4x**, from **14.5%** to **34.5%**.
  - Improved recall **6.7** times from **14%** to **93.6%** by sourcing new impersonator-candidate signals, reducing victim prevalence by **12%** of the team-wide goal of **30%**.
- **Partnered with cross-functional ML infra teams and prompt engineers** to enable multi-media inputs for Llama 4 bulk inference, build shared datasets and LLM evaluation tooling for rapid prompt iteration, and automate ablation studies.

Meta; *Software Engineer Intern* — May 2022 - Aug 2022

- **Reduced demographic bias in a face-verification deep learning model** for Meta Reality Labs
- **Researched and implemented state of the art loss functions and convolutional neural network architectures**, and increased performance from **95.0%** to **98.2%** with a model **five** times smaller than the one previously used.
- **Built an internal analysis toolbox** using PyTorch Lightning, TensorBoard, and Matplotlib to automate bias-metric visualizations and facial-embedding interpretability (Grad-CAM, t-SNE, PCA, LDA).

Birch.ai; *AI Researcher Intern* — Sept 2021 - Nov 2021

- **Built a synthetic data pipeline** that generated documents with tabular content and extracted structured data for model training.
- **Trained computer vision models** using the PyTorch-based MMDetection toolbox to detect tables and signature blocks in healthcare documents.

Intuit; *Software Engineer Intern* — May 2021 - Aug 2021

- **Contributed to Intuit's tax knowledge engine** by applying calculation and completeness graphs for knowledge engineering.
- **Built a version-control system for knowledge-engineering artifacts** using React, Node.js, and Express with GitHub integration.
- **Implemented a custom version of Myers' diff algorithm** to auto-detect writing syntax and generate personalized visual diffs of tabular data across three levels of granularity.

PROJECTS

Monkey Pose Estimation — *Python, Pytorch, PyTorch Lightning, Hydra*

- Built a straightforward and powerful deep learning module using the **PyTorch Lightning** framework for organizing datasets and models and **Hydra** for configuring parameters.
- Trained state-of-the-art deep learning models for the pose estimation task, including HRNet and convolutional pose machines, on the OpenMonkeyChallenge dataset.
- Achieved a mean per joint position error (MJMPE) score of **0.071**, probability of correct keypoint (PCK) score of **0.991**, and mean average precision (mAP) score of **0.786**.

SKILLS

Programming Languages: Python, Java, SQL, Matlab, HTML, CSS, JavaScript, TypeScript
Frameworks/Technologies: ReactJS, NodeJS, Flask, FastAPI Pytorch, Pandas, Git, Latex

EDUCATION

University of Minnesota-Twin Cities — *Aug 2024*
Bachelor of Science in Computer Science and Mathematics — GPA: 3.8/4.0

AWARDS

1st Place Winner — Minnehack 2021 Hackathon — January 2021
Top Harmony Hack — LionHACK2022 Hackathon — April 2022
Ava Labs 3rd Place Winner — LionHACK2023 Hackathon — April 2023
AIME Qualifier — 2018